

PROPOSAL

Tittle = Real Time Phishing Attack Detector

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Abstract

Phishing attacks pose a significant threat in today's digital landscape, causing individuals to lose money daily. Current statistics indicate that approximately 3.4 billion phishing emails are sent each day, making phishing the most widespread form of cybercrime.

One of the primary reasons phishing attacks have escalated to such high levels is the lack of awareness and education about these threats. Many individuals do not fully understand what phishing is or how to identify fraudulent emails. Additionally, cybercriminals now leverage artificial intelligence to craft highly sophisticated and convincing scam emails, making them more deceptive than ever before.

My motivation for developing this system stems from my admiration for the *Scammer Payback* YouTube channel. Their efforts in exposing scammers, hacking their networks, revealing their identities, and reclaiming stolen funds for victims are truly inspiring. However, given that 3.4 billion phishing emails are sent daily, it is impractical for a single group to counter every attack.

Therefore, I decided to create a system designed to help non-technical individuals recognize and avoid phishing attacks, ultimately providing them with the knowledge and tools to protect themselves from cyber threats.

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1)Introduction

Subject Area

Phishing attacks exploit human vulnerabilities to gain unauthorized access to sensitive information (Hong, 2012). With email being a primary attack vector, traditional detection methods like rule-based filters are ineffective against evolving threats (Abutair et al., 2020). Deep learning offers a promising solution by analyzing linguistic and structural patterns to detect phishing emails (Zhang et al., 2021).

Background of the Scenario

Over 90% of social engineering breaches originate from phishing emails (Verizon, 2022). As attackers refine their tactics, machine learning models have proven more effective than signature-based detection (Gupta et al., 2018). This study develops a real-time phishing email monitoring tool, integrating a deep learning model with a flask web application for immediate threat detection and notifications.

Requirement and Importance of Research

The research aims to create an automated phishing detection system with:

- Continuous email monitoring via IMAP/SMTP
- Real-time phishing classification using deep learning
- Instant user notifications upon threat detection
- Easy deployment through Flask
- Desktop application using Tkinter

Traditional anti-phishing measures struggle with zero-day attacks (Kumar & Kumar, 2020). By leveraging deep learning, this system enhances cybersecurity resilience and adaptability. Cloud-based integration ensures scalability for individuals and organizations (Al-Hamar et al., 2021).

References

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2)Research Questions and Project Objectives

Research Questions

This research aims to address the following key questions:

- How effective are deep learning models in detecting phishing emails compared to traditional rule-based methods?
- How to Find datasets
- How to access the Realtime email
- What features (e.g., email content, sender metadata, embedded links) contribute the most to accurately classifying phishing emails?
- How can real-time monitoring and notification systems enhance phishing email detection for end users?
- What are the challenges in implementing a scalable and user-friendly phishing detection tool within a flask web application?

By answering these research questions, the study seeks to evaluate the feasibility, accuracy, and efficiency of integrating deep learning models into real-time phishing detection systems.

Project Objectives

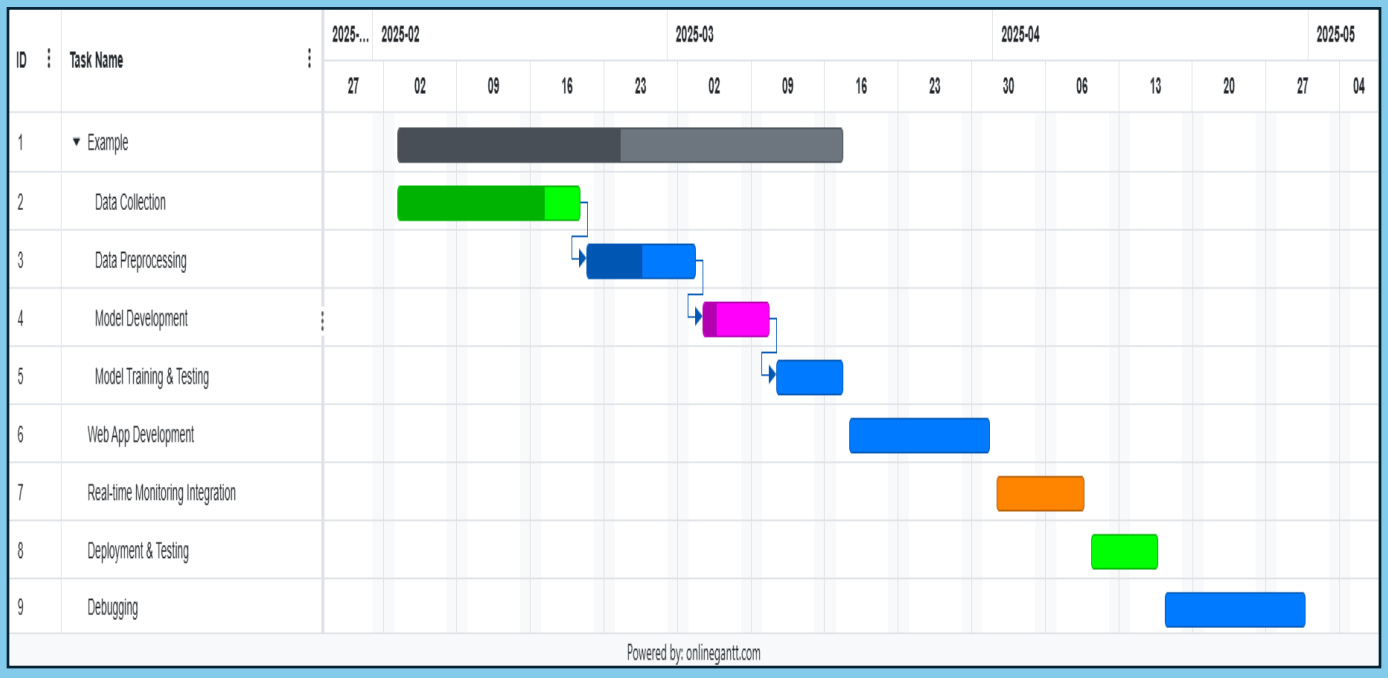
The main objectives of this research project are:

1. **Develop a deep learning model for phishing email detection:**
    - Train and evaluate a neural network-based model to classify phishing and legitimate emails.
    - Optimize the model for high accuracy and minimal false positives.
2. **How to Find datasets**
   Using Kaggle open datasets. Find as much as datasets. To combine those datasets and feed as much as possible.

3. **Integrate the model into a real-time email monitoring system:**
    - Implement IMAP protocols for continuous email scanning.
    - Ensure seamless detection of phishing attempts as emails arrive in users' inboxes.

4. **Provide real-time notifications to users upon detection of phishing emails:**
    - Design a notification system that alerts users via web and mobile interfaces.
    - Ensure that alerts contain relevant information and recommended actions.
5. **Deploy the phishing detection tool within a flask web application:**
    - Develop a user-friendly web interface where users can download and configure the monitoring tool.
    - Ensure secure authentication and efficient performance of the application.
6. **Evaluate the effectiveness of the proposed system:**
    - Test the system against benchmark datasets and real-world email samples.
    - Assess user feedback to improve usability and detection accuracy.

3)Guntt chart



4)Project feasibility

1. Technical Feasibility

1.1 Technical Resource Analysis

The project requires hardware, software, and cloud resources for model training and real-time monitoring.

Resource	Specification	Procurement Capability
Server/Cloud Hosting	AWS/GCP/Azure (GPU-enabled)	Subscription-based
Deep Learning Model	TensorFlow for phishing detection	Open-source
Email API Access	Gmail API, Microsoft Graph API	Free & paid tiers
Database	PostgreSQL/MySQL for email logs	Open-source
Frontend Framework	React.js/Vue.js	Open-source

1.2 Technical Competence Feasibility

Developing and maintaining this project requires expertise in multiple domains.

Competency	Requirement
Deep Learning	Training & deploying phishing detection models
Cybersecurity	Implementing secure email monitoring and data handling
Web Development	Developing Flask backend & React-based UI
Cloud Computing	Managing cloud services for real-time processing
Competency	Requirement

2. Economic Feasibility

Estimated Costs

Cost Component	Estimated Cost (Annual)
Cloud Hosting (AWS/GCP)	$50 - $100
Email API Access (Premium)	$20 - $100
AI Model Training (GPU)	$50 - $300
Web Application Hosting	$100 - $500
Security & Compliance	$500 - $1,50
Total Estimated Cost	$720 - $2000

3. Operational Feasibility

Key Considerations

Aspect	Description
Security Compliance	Adheres to GDPR, CCPA, and ISO 27001 for email data handling
User Privacy	No email content stored, only phishing risk scores recorded
Legal Approvals	Compliance with email providers' terms (Google, Microsoft)
Deployment Strategy	Available as downloadable software & cloud service

5)Ethical Considerations

The phishing email detection system primarily involves the **analysis of email content**, which may contain **sensitive user data**. Therefore, ethical considerations are crucial to ensure compliance with **data protection regulations** such as **GDPR and CCPA**.

Since this research does not involve **human participants, animals, or medical trials**, it is classified as a **low-risk project**. However, due to the handling of potentially **confidential emails**, an **ethics review** was conducted to ensure that no personally identifiable information (PII) is stored or misused.

The research follows **strict ethical guidelines**, including:

- **Informed Consent:** Users will be informed about data collection and usage policies before installing the tool.
- **Data Anonymization:** Email content will be processed using **hashing and encryption techniques** to protect privacy.
- **Secure Storage & Access:** All email data is stored securely, with access restricted to authorized personnel only.
- **Compliance with Legal Standards:** The system adheres to **GDPR Article 5 (data minimization and security)** and **CCPA guidelines** for handling user data.

An ethics approval request was submitted to the **Institutional Ethics Committee**, and approval was granted under the condition that **no personal data is stored, shared, or used for unintended purposes**. Continuous compliance monitoring will be conducted to **ensure adherence to ethical and legal requirements**.

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